higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATES (VOCATIONAL)

ENGINEERING TECHNOLOGY
NQF LEVEL 2

13 NOVEMBER 2019

This marking guideline consists of 5 pages.

Copyright reserved

SECTION A

QUESTION 1: GENERAL

1.1	1.1.1	C	
	1.1.2	B	
	1.1.3	A or B	
	1.1.4	C or D	
	1.1.5	D or C	(5)
1.2	1.2.1	E	
	1.2.2	D	
	1.2.3	A	
	1.2.4	B	
	1.2.5	C	(5)
1.3	1.3.1	T	
	1.3.2	F	
	1.3.3	F	
	1.3.4	T	
	1.3.5	T	(5)
1.4	1.4.1	Hexagon	
	1.4.2	Vernier protractor	
	1.4.3	Clock gauge	
	1.4.4	Bore	
	1.4.5	Ram or cylinder	
		(5 × 1)	(5)
			[20]

TOTAL SECTION A: 20

SECTION B

QUESTION 2: ENGINEERING HAND TOOLS

2.1
- Hammers
- Pliers
- Spanners
- Chisels
- Allen keys

(Any other relevant tools) (5)

2.2
- Type of cut
- Cross-sectional profile
- Length
- Degree of coarseness

(4)

2.3 •Always use the correct spanner.
•Do NOT use a spanner as a hammer or a lever.
•Do NOT use a pipe to increase the leverage.
•Do NOT hammer on a spanner to loosen or tighten it.
•Always pull on the spanner. (Any 4 × 1) (4)

2.4 •Incorrect and careless use
•Blade too soft
•Blade too hard
•Wrong blade type
•Too coarse a pitch (5)

2.5 It is used to drive out pins,✓ for example, a cotter pin on a bicycle.✓
(Any other example) (2)
[20]

QUESTION 3: ENGINEERING POWER TOOLS

3.1

POWER TOOLS	USE IN THE INDUSTRY
Portable electric hand drill✓	Drilling where a work piece cannot fit into a pedestal drill✓
Pencil grinder✓	Grinding delicate work pieces✓
Air compressor✓	Generating air pressure for various proposes✓
Hydraulic press✓	Fitting and removing of bearings, etc✓
Band saw✓	Cutting where a hacksaw would take too long✓
Portable angle grinder✓	Grinding objects that cannot fit on to a bench grinder✓

(Any other relevant tools and their uses – 2 × 6) (12)

3.2 3.2.1 It is used to remove solids.
3.2.2 It is used to remove water vapour.
(2 × 1) (2)

3.3 3.3.1 •Check all pins on the ram for wear and tear.
•Check that the yoke is not bent. (2)

3.3.2 • Check for leakage in the line.
• Check the oil levels. (2)

3.3.3 • Check the electrical plug for damage.
• Check the casing for cracks. (Any ONE) (1)

3.3.4 • Check the casing for cracks.
• Check the electrical plug. (Any ONE) (1)

3.4 3.4.1 Metal grinding disc – used on metal only
3.4.2 Masonry disc – used for cutting masonry only
(2 × 1) (2)
[22]

QUESTION 4: ENGINEERING MEASURING EQUIPMENT AND SI UNITS

4.1
- Vernier callipers
- Outside callipers
- Depth micrometer
- Dial test indicator
- Internal micrometer (Any other relevant instrument) (5)

4.2 4.2.1 To measure the thickness of a shaft, or the size or depth of a hole
4.2.2 To measure the size of a shaft
4.2.3 To measure the bore of a cylinder
4.2.4 To measure the end-float of a shaft in a machine
(Any other relevant answers – 4 × 1) (4)

4.3 12 min = 12 × 60 = 720 seconds✓
1 hour = 1 × 60 min = 60 × 60 = 3 600 seconds✓
56 min = 56 × 60 = 3 360 seconds✓
Total time = 720 + 30 + 3 600 + 3 360 + 42 = 7 752 seconds✓
Total time = 7 752 / 60 = 129 minutes, 12 seconds✓ (5)

OR

4.3 12 min + 56 min + 60 min = 128 minutes✓
30 sec + 42 sec = 72 seconds✓
72 sec/60 sec = 1 min 12 seconds✓
128 min + 1 min 12 sec = 129 minutes, 12 seconds✓✓ (5)

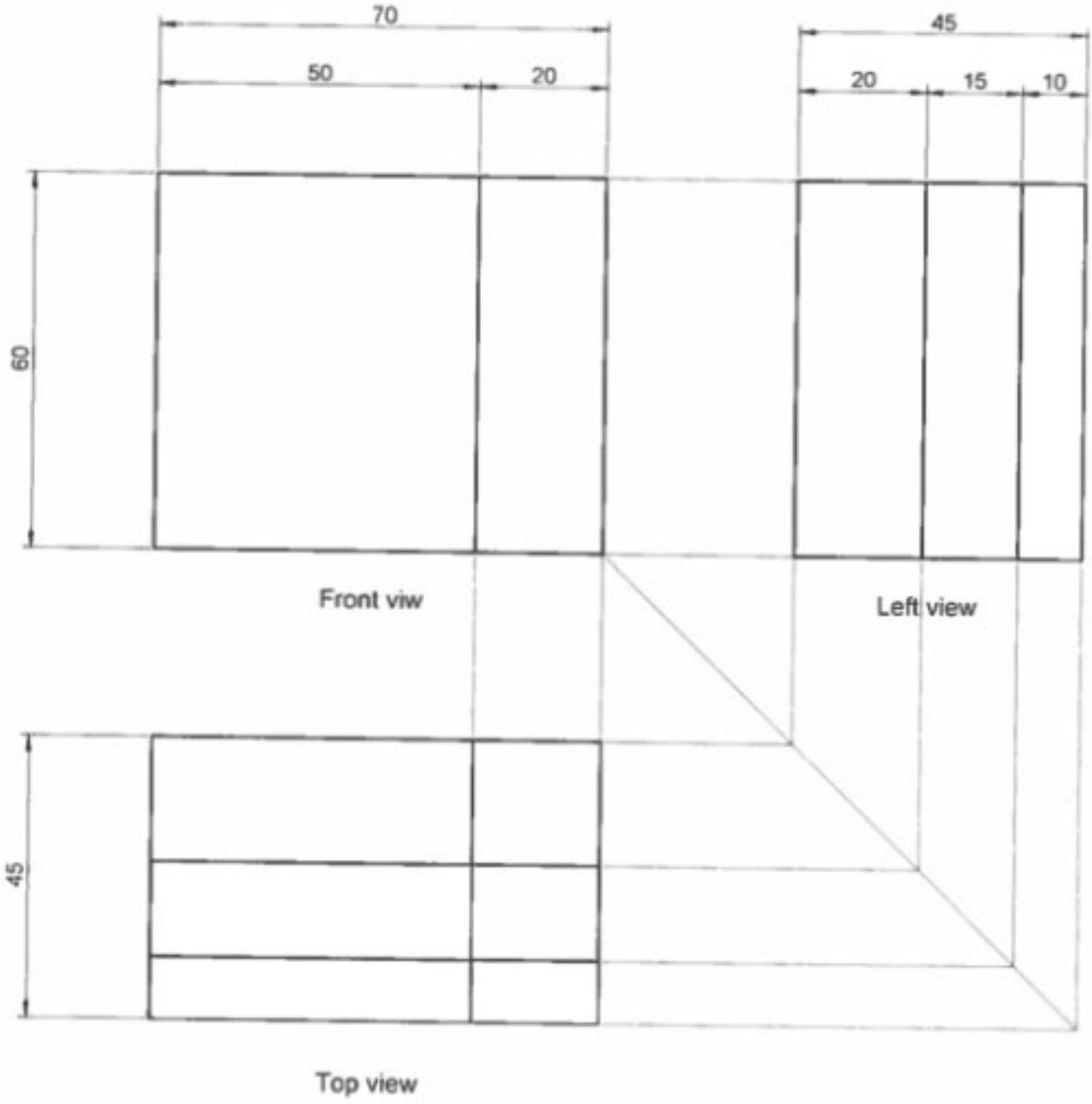
4.4
- Do NOT use the Vernier blades as a lever.
- Do NOT drop or bump the Vernier protractor.
- Keep the Vernier protractor clean and well oiled.
- Store the Vernier protractor in a protected case after use. (4)

4.5
- Inspect the slip gauge for any surface damage.
- Use a proper solvent to clean the gauge blocks thoroughly.
- Apply a small amount of oil to the measuring surface.
- Use clean lens paper, only leaving a thin layer of oil.
- The thin layer of oil will ensure satisfactory wringing. (5)

[23]

Copyright reserved

QUESTION 5: BASIC ENGINEERING DRAWING



[15]

TOTAL SECTION B: 80
GRAND TOTAL: 100

Copyright reserved